Proceedings Track

Truth Has a Boundary. A Machine-Verified Topological Law for Models That Separate Truth from Falsehood

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Abstract

Probing studies show that truth is easy to read off a language model's activations. Linear probes find truth directions. True and false statements separate in representation space. We ask the reverse question. What does the shape of that space force on any model that answers over it? The model is any continuous map from a connected representation space to answers and confidences. Truth is a continuous signed field with the truth boundary as its zero set. Our headline result is a coupling law. If the model ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query is forced to carry confidence exactly at the decision threshold while its answer sits exactly on the truth boundary. The law mentions no safety guarantee, so it cannot be dissolved by adjusting how a guarantee treats its own threshold. Consequences follow per policy. A guarantee that includes its threshold is impossible. A guarantee that excludes its threshold survives but is silent at the forced point. Under Lipschitz regularity the forced point widens into an ambiguity tube of computable width, and any feasible system must carry strictly positive truth-side slack. For nonzero linear probes the truth boundary is exactly an affine hyperplane of codimension one. We further prove a symmetry version that needs no coverage assumption, a discrete analysis that prices the removal of topology, and multi-turn and stochastic extensions. The stochastic dichotomy is where genuine falsehood appears. Every theorem in this paper is machine-verified in Lean 4 against Mathlib, with no sorry, and the headline results reduce to the three standard kernel axioms. An appendix observes the predicted coupling in three sub-billion-parameter language models, including with the model's own confidence signal and against null baselines.

Keywords: representational geometry, topology, uncertainty, calibration, intermediate value theorem, Borsuk-Ulam, formal verification, Lean

1. Introduction

Truth is geometrically simple inside language models. Linear probes separate true from false statements in activation space (Marks and Tegmark, 2024; Azaria and Mitchell, 2023). Latent knowledge can be found without supervision (Burns et al., 2023). Steering activations along learned directions makes models more truthful (Li et al., 2023; Zou et al., 2023). All of this work treats truth as a continuous, decodable field over representation space.

We take that picture seriously as geometry and ask what follows. Suppose truth is a continuous scalar field on a connected representation space, and the model's answers and confidences are continuous fields on the same space. These assumptions mention no architecture, training method, or scale. They apply to any model for which the fields are defined and continuous over the stated domain. What they force is a coupling law.

> If the model ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query carries confidence exactly at the threshold while its answer sits exactly on the truth boundary.

The proof is short and we do not pretend otherwise. The value of the law lies in what it does not mention. It contains no safety guarantee and no convention about thresholds, so every policy question becomes a corollary. An inclusive guarantee is impossible. An exclusive guarantee survives but says nothing at the one point topology guarantees to exist. Slack buys safety, and we compute exactly how much. Under Lipschitz regularity the forced point widens into an ambiguity tube of computable width. For the linear probes used in practice, the truth boundary is literally an affine hyperplane.

Contributions.

1. **A boundary coupling law** (Section 4). Coverage plus continuity force an exact truth-boundary point. Truth separation forces threshold confidence at such a point. The combined law is convention-free, and a taxonomy of corollaries then settles each policy.

2. **The shape of the boundary** (Section 5). The truth boundary is closed. Every continuous path from a true answer to a false answer crosses it. For a nonzero linear probe it is an affine translate of the probe's kernel, of dimension exactly $d - 1$. The title of this paper is meant literally.

3. **The ambiguity tube** (Section 6). With Lipschitz confidence, decisive regions are separated by a corridor of width at least $2\gamma/L$. Around the forced point, truth margin and confidence margin are bounded by the Lipschitz data. Any system with band separation and a threshold-inclusive guarantee must have strictly positive truth-side slack.

4. **A symmetry version** (Section 7). If semantic negation acts freely and continuously and the truth field is odd under it, a boundary point is forced with no coverage assumption at all.

5. **A discrete analysis** (Section 8) that prices the removal of topology, and **extensions** (Section 9) to multi-turn and stochastic models.

6. **Full machine verification** (Section 10). Every theorem in this paper corresponds to a named, sorry-free Lean 4 result checked against Mathlib.

What the contribution is. The mathematics is deliberately elementary, and the contribution is not technical difficulty. It is a framing in which forced uncertainty is a topological property of representation space, stated in the vocabulary of the probing literature. It is precision, since the theorems name the exact assumptions in play, including the threshold convention other treatments leave implicit, so debate moves from proofs to premises. And it is verification, since every statement is machine-checked and the premises are the only thing left to argue about.

What the theorems do not say. They do not say models err often, and they assign no probability to failure. The forced point of the deterministic results lies on the truth boundary, neither strictly true nor strictly false. Derived falsehood appears only in the stochastic dichotomy.

2. Related work

Hallucination inevitability. Kalai and Vempala (2024) show statistically that calibrated models must hallucinate on facts seen once in training, and Kalai et al. (2025) extend this line. Xu et al. (2024) give a computability-theoretic inevitability argument. Our mechanism is topological, lives in representation space, and localizes the forced behavior at the truth boundary. It is also more modest, since the deterministic theorems force uncertainty at the boundary rather than confident falsehood.

Geometry of truth in representations. Linearly decodable truth (Marks and Tegmark, 2024), unsupervised latent-knowledge discovery (Burns et al., 2023), lie detection from internal states (Azaria and Mitchell, 2023), and causal steering along truth directions (Li et al., 2023; Zou et al., 2023) all model truth as a continuous decodable field on activation space. That is exactly our hypothesis on δ, and confidence signals (Farquhar et al., 2024; Kadavath et al., 2022) are treated similarly. Our results give this program a frontier, since a probe that exactly separates truth over a connected domain with both truth values realized must contain an exact-uncertainty point on its own boundary.

Topology, geometry, and calibration. The manifold hypothesis (Fefferman et al., 2016) and the measured geometry of embedding spaces (Cai et al., 2021) motivate connected-domain modeling. Topological analyses of networks (Naitzat et al., 2020; Hensel et al., 2021) and geometric deep learning (Bronstein et al., 2021) engineer or analyze such structure. We derive constraints from it. The symmetry section is a one-dimensional relative of the Borsuk-Ulam theorem (Borsuk, 1933; Matoušek, 2003). Our separation condition is not statistical calibration in the sense of Guo et al. (2017), and Remark 6 spells out the difference.

Formalized machine learning theory. All claims are verified in Lean 4 (de Moura and Ullrich, 2021) against Mathlib (mathlib Community, 2020). To our knowledge this is the first fully machine-verified impossibility-style result on model truthfulness stated at the level of representation geometry.

3. The setting

$\mathcal{X}$ is a topological space of queries as the model represents them. Think of prompt embeddings or residual-stream states. $\mathcal{A}$ is a topological space of answers. Our standing assumption is simple.

> **(T)** $\mathcal{X}$ is connected and nonempty.

Assumption (T) deserves care. Ambient embedding spaces are standardly modeled as $\mathbb{R}^d$ or connected submanifolds (Fefferman et al., 2016; Cai et al., 2021), which satisfy (T). But the set of representations a token-driven model actually reaches may be disconnected. When guarantees range over the ambient space, as they implicitly do whenever a probe or steering vector is applied at arbitrary points, (T) holds. Section 8 analyzes what remains without it.

Definition 1 (Model field) *A* model *is a map $F = (a, c)$ from $\mathcal{X}$ to $\mathcal{A} \times \mathbb{R}$. Here $a(x)$ is the answer at representation x and $c(x)$ is its scalar confidence. F is* continuous *if both component maps are.*

Definition 2 (Truth field) *A* truth field *is a continuous map δ from $\mathcal{X} \times \mathcal{A}$ to $\mathbb{R}$, read as a signed truth-distance. Negative means the answer is strictly true. Positive means strictly false. Zero is the* truth boundary. *The* realized truth field *of a model is $\delta_F(x) = \delta(x, a(x))$.*

A linear probe $w^\top x - b$ on activations (Marks and Tegmark, 2024; Burns et al., 2023) is a candidate continuous δ_F, and its decision boundary is the truth boundary. We note the gap honestly. Probing experiments fit classifiers on sampled representations, and the zero set of a probe is the probe's classification boundary. Whether it coincides with semantic truth away from those samples is an empirical premise, not a theorem. Our results hold under that premise.

Definition 3 (Truth-separating at threshold τ) *Fix a threshold τ, a truth slack $\varepsilon \geq 0$, and a confidence margin $\gamma \geq 0$. F is* (ε, γ)-separating at τ *if for all x two implications hold. If $\delta_F(x) < -\varepsilon$ then $c(x) > \tau + \gamma$. If $\delta_F(x) > \varepsilon$ then $c(x) < \tau - \gamma$. The truth slack is measured in units of δ_F and the margin in units of c, so the condition survives independent rescaling of either field. We say* exactly separating *when both are zero.*

Definition 4 (Covering) *F is* ε-covering *if some query has $\delta_F < -\varepsilon$ and some query has $\delta_F > \varepsilon$. At $\varepsilon = 0$ this says the model is somewhere strictly right and somewhere strictly wrong.*

Definition 5 (Guarantees) *A* threshold-inclusive guarantee *says that $c(x) \geq \tau$ implies $\delta_F(x) < 0$. A* threshold-exclusive guarantee *says that $c(x) > \tau$ implies $\delta_F(x) < 0$.*

Covering is the weakest condition, since any deployed model that is ever right and ever wrong satisfies it. Exact separation idealizes a perfect truth probe used as the confidence signal, and the guarantees idealize hallucination mitigation. The coupling results hold at any threshold, and the biconditional variants at one half.

Remark 6 (This is not statistical calibration) *Statistical calibration asks that confidence match accuracy on average (Guo et al., 2017). Separation is pointwise and much stronger, since confidence classifies truth, and our impossibility results target this stronger object. Separation is also one-way by construction. The coupling law and its corollaries use only the two truth-to-confidence implications, while the symmetry and discrete results use the biconditional variant, which implies them (Lean `strictCalibrated_to_epsCalibrated_zero_half`).*

Remark 7 (On continuity) *Continuity idealizes the smoothness of trained networks. Softmax confidences and embedding maps are continuous by construction, while hard argmax decoding is not. Dropping continuity is a genuine escape, and Section 8 prices it.*

4. The boundary coupling law

Topology enters through one theorem, which needs no confidence at all.

Theorem 8 (Boundary existence) *Let $\mathcal{X}$ satisfy (T). Let the answer map and the truth field be continuous, and let F be 0-covering. Then some x_0 has $\delta_F(x_0) = 0$.*

Proof The realized truth field δ_F is continuous as a composite. It is negative somewhere and positive somewhere. The intermediate value theorem on the connected space $\mathcal{X}$ gives an exact zero. This is Lean theorem `model_truth_boundary_nonempty`, and the IVT step is Mathlib's `IsPreconnected.intermediate_value₂`. ■

Confidence enters next, and the headline law follows.

Theorem 9 (Boundary coupling) *Let $\mathcal{X}$ satisfy (T). Let c be continuous, and let F be 0-covering and exactly separating at τ. Then some x_0 satisfies both*

$$c(x_0) = \tau \qquad \text{and} \qquad \delta_F(x_0) = 0.$$

Proof Separation converts the coverage witnesses into confidence witnesses above and below τ. The IVT gives x_0 with $c(x_0) = \tau$. If $\delta_F(x_0)$ were negative, separation would force $c(x_0) > \tau$. If it were positive, separation would force $c(x_0) < \tau$. Both fail, so $\delta_F(x_0) = 0$. This is Lean theorem `boundary_coupling`. ■

The law mentions no guarantee, so no threshold convention can dissolve it. The model is forced into total uncertainty at exactly the place where truth changes sign. What that point means for safety is a matter of policy, and each policy is a corollary.

Corollary 10 (Inclusive guarantees are impossible) *Add a threshold-inclusive guarantee to the hypotheses of Theorem 9. Then the hypotheses are jointly contradictory. This is the trilemma.*

Corollary 10 is Lean theorem `closed_guarantee_impossible`, and its fixed-threshold biconditional version is `hallucination_trilemma`.

Corollary 11 (Exclusive guarantees go silent) *Add a threshold-exclusive guarantee instead. No contradiction follows. Instead the forced point satisfies $c(x_0) = \tau$, so the guarantee's antecedent fails exactly there. The system owns a query about which its guarantee says nothing, and that query sits on the truth boundary. This is Lean theorem `open_guarantee_silent`.*

Remark 12 (The convention question, settled) *A natural objection runs as follows. Define the guarantee with a strict inequality and the contradiction disappears, so the impossibility is an artifact of a convention. The taxonomy above is our answer. The coupling law is convention-free. The convention only selects which corollary describes the forced point. Under an inclusive guarantee it is a contradiction. Under an exclusive guarantee it is a point of forced silence. Either way the system cannot be decisive with a guarantee everywhere, and the forced point exists regardless.*

5. The shape of the boundary

The forced set is not an isolated curiosity. It has structure, and for the probes used in practice it is literally a hyperplane.

Theorem 13 (The boundary is closed) *For any topological $\mathcal{X}$ and $\mathcal{A}$, continuous δ and answer map, the set $\{x \mid \delta_F(x) = 0\}$ is closed in $\mathcal{X}$. This is Lean theorem `model_truth_boundary_isClosed`.*

Theorem 14 (Every path crosses) *Let p be a continuous path in $\mathcal{X}$ from a query where the model answers strictly truly to a query where it answers strictly falsely. Then some point on the path lies exactly on the truth boundary. This is Lean theorem `path_crosses_truth_boundary`.*

The boundary therefore separates the true region from the false region, and no continuous route between them avoids it. For linear probes we can say exactly what it is.

Theorem 15 (Linear probes have hyperplane boundaries) *Let f be a nonzero linear probe on a d-dimensional real space, with offset b. Then the boundary set $\{x \mid f(x) = b\}$ is an affine translate of the kernel of f, and that kernel has dimension exactly $d-1$. These are Lean theorems `linear_probe_boundary_coset` and `linear_probe_boundary_dim`, the latter by rank-nullity.*

For the linear probes of the truth-direction literature, the truth boundary is therefore a codimension-one affine hyperplane through representation space, and the coupling law places a threshold-confidence answer somewhere on it. Truth has not merely a boundary point but a boundary surface, and the confidence threshold must meet it.

6. The ambiguity tube

Exact separation is an idealization, so we quantify what slack changes. Two Lipschitz facts first. They need only a metric space.

Theorem 16 (Corridor width) *Let c be L-Lipschitz with $L > 0$ and let $\gamma > 0$. Then any query with $c \geq \tau + \gamma$ and any query with $c \leq \tau - \gamma$ are at distance at least $2\gamma/L$. This is Lean theorem `confidence_gap_width`.*

Theorem 17 (Ambiguity tube) *Let c be L_c-Lipschitz and δ_F be L_δ-Lipschitz, and let x_0 be a coupled point from Theorem 9. Then every query within distance r of x_0 has truth margin at most $L_\delta r$ and confidence within $L_c r$ of the threshold. This is Lean theorem `ambiguity_tube`, with the margin bound alone as `truth_margin_tube`.*

The forced point is therefore not one pathological query. Around it sits a tube of near-ambiguous queries controlled by the two Lipschitz constants, and decisive regions cannot approach closer than the corridor width. Slack changes what the forced point looks like, and we can say exactly how.

Theorem 18 (Approximate coupling) *Let $\mathcal{X}$ satisfy (T), let c be continuous, and let F be ε-covering, (ε, γ)-separating at τ, and carry a threshold-inclusive guarantee. Then some x_0 has $c(x_0) = \tau$ and $-\varepsilon \leq \delta_F(x_0) < 0$. This is Lean theorem `two_slack_approx_coupling`, with the exact case as `exact_from_approx`.*

Theorem 19 (Truth slack must be positive) *Under the hypotheses of Theorem 18, the truth slack satisfies $\varepsilon > 0$ for every margin $\gamma \geq 0$. Zero truth slack is infeasible. This is Lean theorem* `truth_slack_must_be_positive`.

Slack is therefore a genuine geometric budget, and positivity attaches to the truth side alone, whatever the confidence margin. We propose reading the infimum of feasible truth slack as the system's truth-boundary width. The corridor is also clean. Under the guarantee, no query with confidence between τ and $\tau + \gamma$ sits exactly on the truth boundary (Lean theorem `no_boundary_in_upper_band_two_slack`).

7. Symmetry replaces coverage

The coupling law uses coverage to seed the IVT. Symmetry can replace coverage entirely. Suppose semantic negation acts on $\mathcal{X}$ as a continuous involution σ with no fixed point, and suppose the realized truth field is odd under it. Negating the query then flips the sign of the truth-distance. Oddness idealizes the observation that negation approximately reflects truth directions in activation space (Marks and Tegmark, 2024; Burns et al., 2023).

Theorem 20 (Odd fields vanish) *Let $\mathcal{X}$ satisfy (T) with such an action σ, and let g be continuous and odd under σ. Then g has a zero. This is Lean theorem* `antipodal_odd_has_zero`.

Proof Pick any x. If $g(x) = 0$ we are done. Otherwise $g(x)$ and $g(\sigma x) = -g(x)$ have opposite signs, and the IVT on the connected space yields a zero of g. ∎

This is the scalar shadow of the Borsuk-Ulam theorem (Borsuk, 1933; Matoušek, 2003). It needs no sphere. Any connected space with a free continuous $\mathbb{Z}/2$ action supports it.

Theorem 21 (Antipodal coupling) *Let $\mathcal{X}$ satisfy (T) with such an action, let F be continuous with biconditional separation at $\frac{1}{2}$, and let δ_F be odd. Then some x has $c(x) = \frac{1}{2}$ and $\delta_F(x) = 0$. This is Lean theorem* `antipodal_hallucination_trilemma`.

No coverage hypothesis appears. A model answering a negation-closed family of questions is forced onto the truth boundary by symmetry alone. The moral is pointed for this community, since equivariance is usually a design virtue (Bronstein et al., 2021) and here it is the engine of the obstruction.

8. What leaving the continuum costs

Token spaces are finite, and one may object that connectedness does illegitimate work. The discrete analysis makes the objection and its price exact. Without topology the boundary witness cannot be derived. It must be assumed.

Theorem 22 (Discrete impossibility) *Let $\mathcal{X}$ and $\mathcal{A}$ be arbitrary sets with arbitrary F and δ. Suppose F has biconditional separation at $\frac{1}{2}$ and some query has $\delta_F = 0$. Then a threshold-inclusive guarantee fails. Moreover, under separation, such a guarantee is possible exactly when no boundary query exists. These are Lean theorems* `boundary_question_impossible` *and* `faithful_iff_boundary_free`.

The boundary witness is assumed here, and it is exactly what connectedness derived for free in Theorem 8. Whether boundary queries exist in a given discrete system is an empirical question. What survives without any witness assumption is a crossing.

Proposition 23 (Discrete sign change) *Let g map $\{0, \ldots, n\}$ to $\mathbb{R}$ with $g(0) < 0 < g(n)$. Then some i has $g(i) < 0 \leq g(i+1)$. This is Lean theorem `discrete_sign_change`.*

Along any token-space path from a true answer to a false one, the realized truth field crosses zero between two adjacent steps. This is weaker than the continuous law, yielding a straddling pair rather than a state on the boundary, and hence no contradiction on its own. The continuum is precisely what upgrades a crossing into a witness.

9. Products and randomness

Multi-turn models. A T-turn model sees a history in $\mathcal{X}^T$. Products of connected spaces are connected, so hypothesis (T) is inherited by history space and the connectedness arguments above apply verbatim to history-dependent models. The Lean artifact transports the trilemma this way (Lean theorem `multi_turn_history_dependent`). Conversation context moves the truth boundary into history space. It does not remove it.

Stochastic models, where falsehood appears. Applied to the expected confidence and truth fields, the coupling law yields a query with expected confidence one half and expected truth-distance zero (Lean theorem `stochastic_coupling_expected`). A dichotomy converts expectation into sampling behavior.

Theorem 24 (Stochastic dichotomy) *Let (Ω, μ) be a probability space and let c, d be real functions on Ω with d integrable and of mean zero. Suppose sample faithfulness holds almost everywhere, meaning $c \geq \frac{1}{2}$ implies $d < 0$, and suppose $\mu\{c \geq \frac{1}{2}\} > 0$. Then $\mu\{d > 0\} > 0$. This is Lean theorem `stochastic_dichotomy`.*

At a forced boundary point, a model that is faithful on almost every sample and confident with positive probability must emit strictly false answers with positive probability. Randomization spreads the boundary across samples rather than eliminating it. This is the one place in the paper where falsehood is derived.

10. Machine verification

Every theorem and proposition in this paper corresponds to a named, sorry-free Lean 4 result (de Moura and Ullrich, 2021) checked against Mathlib (mathlib Community, 2020), both pinned at v4.28.0. The anonymized `HallucinationProofs` artifact ships seventeen source files. Running `lake build` completes with zero errors and no `sorry`. Axiom audits run `#print axioms` on the headline theorems, and each reduces to exactly the three standard kernel axioms. Table 1 in Appendix A maps every result to its identifier. The theorem `hallucination_master_theorem` bundles the core impossibility together with its coverage data. The formal statements are slightly more general than the prose, and the prose claims nothing beyond what is formalized. Some identifiers retain historical names.

11. Discussion

The coupling law rests on connectedness, continuity, coverage, and separation. The trilemma adds a threshold-inclusive guarantee. Each hypothesis can be relaxed, and each relaxation is a design stance. We do not claim this list is exhaustive.

Exclude the threshold from the guarantee. Corollary 11 makes this stance precise. The guarantee survives but owns a boundary point about which it says nothing. Saying so explicitly is healthier than leaving the convention implicit.

Disconnect the space. Abstention breaks the argument at the IVT step. A model that declines to answer on an open region deletes that region from the domain where the fields must interpolate, which can disconnect the true region from the false region. Refusal training, on this reading, is connectedness surgery. Under the symmetry version the cut must separate every representation from its negation image, so piecemeal refusal on isolated hard queries is not enough.

Quantify only over reachable queries. The forced point may be an ambient activation that no real prompt produces, and a guarantee quantified only over reachable representations is then untouched. The objection is fair, and it is the domain version of disconnecting the space. But probing and steering pipelines apply their readouts at interpolated and edited activations, which is deployment over the ambient space, and a guarantee over the uncertified reachable set inherits every unknown of that set. Restricting the domain is a coherent design stance, and it should be stated as one.

Give up coverage or continuity. A model that never answers strictly falsely escapes. That is vacuous for general-purpose systems and meaningful for retrieval-grounded ones restricted to verified content. Dropping continuity also escapes, at the price computed in Section 8, where only the adjacent-crossing statement survives unconditionally.

Separate to a band. Theorems 18 and 19 make slack the correct optimization target. A feasible system must carry strictly positive truth slack, with guarantees available only outside the band between τ and $\tau + \gamma$. The infimum of feasible truth slack is measurable, and we propose it as a diagnostic.

Implications for probing and steering. Truth-probe pipelines posit a continuous δ_F, and steering methods move representations relative to its zero set (Li et al., 2023; Zou et al., 2023). Our results say that completing such a pipeline over a connected domain, with both truth values realized, forces an exact-uncertainty point on the probe's own boundary. For linear probes that boundary is a hyperplane, and mitigation effort should go there.

Limitations and outlook. The results are existence statements, so nothing follows about error rates. Deterministic conclusions concern boundary points rather than confident falsehood, with Theorem 24 as the stochastic exception. The truth field idealizes probe behavior, and exact oddness idealizes approximate negation symmetry. Appendix C reports a first empirical check on three sub-billion-parameter models. Natural next steps are higher Borsuk-Ulam machinery for jointly odd semantic axes and regular-value descriptions of nonlinear probe boundaries. The geometry of representation space can prove theorems, and machines can check them.

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Appendix A. Formal statement conventions

The Lean formalization states results over arbitrary topological spaces `Q` and `A`, with a `ConnectedSpace` instance on `Q`, a model `M` from `Q` to `A x R`, and a truth field `delta` on `Q x A` (rendered here in ASCII). The realized truth field is

```
fun q => delta (q, (M q).1)
```

and confidence is `(M q).2`. The main conditions are named as follows.

- Separation at threshold `c` with truth slack `eps` and confidence margin `gamma` is `TwoSlackSeparating M delta c eps gamma`. The equal-slack case is `EpsCalibrated M delta c eps`, definitionally (`epsCalibrated_iff_twoSlack`).
- Covering with slack `eps` is `EpsCovering M delta eps`.
- The biconditional variant at threshold one half is `StrictCalibrated M delta`.

- The threshold-inclusive guarantee at threshold one half is `TrilemmaFaithful M delta`. The general threshold version appears inline in the theorems that use it.

The embedding-space instantiation uses `EuclideanSpace R (Fin d)`. The quantitative results use `PseudoMetricSpace` and explicit Lipschitz hypotheses, and the hyperplane results use `LinearMap` with rank-nullity.

Table 1: Every result in the paper and its machine-checked counterpart.

Result	Lean identifier
Thm. 8	`model_truth_boundary_nonempty`
Thm. 9	`boundary_coupling`
Cor. 10	`closed_guarantee_impossible`, `hallucination_trilemma`
Cor. 11	`open_guarantee_silent`
Thm. 13	`model_truth_boundary_isClosed`
Thm. 14	`path_crosses_truth_boundary`
Thm. 15	`linear_probe_boundary_coset`, `linear_probe_boundary_dim`
Thm. 16	`confidence_gap_width`
Thm. 17	`ambiguity_tube`, `truth_margin_tube`
Thm. 18	`two_slack_approx_coupling`, `exact_from_approx`
Thm. 19	`truth_slack_must_be_positive`
Upper-band exclusion	`no_boundary_in_upper_band_two_slack`
Thm. 20	`antipodal_odd_has_zero`
Thm. 21	`antipodal_hallucination_trilemma`
Thm. 22	`boundary_question_impossible`, `faithful_iff_boundary_free`
Prop. 23	`discrete_sign_change`
Multi-turn	`multi_turn_history_dependent`
Stochastic expected-value coupling	`stochastic_coupling_expected`
Thm. 24	`stochastic_dichotomy`
Bundled master theorem	`hallucination_master_theorem`

Appendix B. Core proof term

The Lean proof of the coupling law, lightly annotated and rendered in ASCII.

```
theorem boundary_coupling
    {Q A : Type*} [TopologicalSpace Q] [ConnectedSpace Q]
    [TopologicalSpace A]
    (M : Q -> A x R) (delta : Q x A -> R) (c : R)
    (hconf : Continuous (fun q => (M q).2))
    (hcov : EpsCovering M delta 0)
    (hcal : EpsCalibrated M delta c 0) :
    exists q0, (M q0).2 = c /\ delta (q0, (M q0).1) = 0 := by
  obtain <<qt, ht>, <qf, hf>> := hcov
  -- separation turns coverage witnesses into confidence witnesses
  have hct : (M qt).2 > c + 0 := hcal.1 qt ht
  have hcf : (M qf).2 < c - 0 := hcal.2 qf hf
  have hct' : (M qt).2 > c := by linarith
  have hcf' : (M qf).2 < c := by linarith
```

```
-- IVT on the confidence field
obtain <q0, _, hq0> :=
  isPreconnected_univ.intermediate_value2
    (mem_univ qf) (mem_univ qt)
    hconf.continuousOn continuous_const.continuousOn
    (le_of_lt hcf') (le_of_lt hct')
refine <q0, hq0, ?_>
-- at conf = c, separation excludes both signs
by_contra hne
rcases lt_or_gt_of_ne hne with hlt | hgt
. have hd : delta (q0, (M q0).1) < -0 := by simpa using hlt
  have := hcal.1 q0 hd
  linarith
. have := hcal.2 q0 hgt
  linarith
```

Appendix C. Empirical illustration on sub-billion-parameter models

The theorems are about idealized fields, so we checked whether their predictions are visible in real models. We used three open language models below one billion parameters. GPT-2 with 124M parameters, Pythia-410M, and Qwen2.5-0.5B. The data are the matched city statements and their negations from the geometry-of-truth datasets of Marks and Tegmark (2024), giving 1496 true and false statement pairs such as “The city of Paris is in France” and its negation.

Setup. We embed each raw statement at its final token and select the layer whose probe has the best validation accuracy. The truth field δ_F is a linear logistic probe trained on one split of cities. The confidence field c is a second linear logistic head trained on a disjoint split with a different seed, so the two fields are independently trained readouts of the same representation space. This is the canonical construction the theory models, since a probing pipeline that uses a truth readout as a confidence signal is exactly a truth-separating c. We then interpolate hidden states between each matched pair, from the true statement to its negation, and record both fields along each path. We keep the paths whose endpoints the probe classifies correctly, which is 83 paths for GPT-2, 104 for Pythia-410M, and 137 for Qwen2.5-0.5B. Test cities are held out from both training splits.

Findings. Table 2 and Figure 2 summarize. First, the premises hold approximately. Linear probes separate truth with accuracy 0.77 to 0.96 even though two of the three models are at chance when asked the question verbally, which replicates the known gap between represented and verbalized truth. Second, every kept path shows the adjacent sign change predicted by the discrete result. Third, confidence is pinned near the threshold at the boundary. The mean value of c at the probe’s zero crossing is 0.52, 0.51, and 0.49 for the three models, against decisive values near 0 and 1 at the endpoints. Fourth, the two independently trained fields couple. The median gap between the zero of δ_F and the threshold crossing of c shrinks as separation quality improves, from 0.25 of the path for GPT-2 to 0.10 for Qwen2.5-0.5B. Fifth, the relative boundary width, the median $|\delta_F|$ over queries whose confidence lies in $[0.4, 0.6]$ divided by the endpoint scale, shrinks from 0.33 for GPT-2 to 0.20 for Qwen2.5-0.5B, though Pythia-410M breaks monotonicity at 0.36. Sixth, the kNN graph of held-out activations places true and false statements in one shared main component for all three models, supporting the connectedness premise on this domain, though two models show a second component and we report this plainly.

The model's own confidence. The design above trains both fields on truth, so nearby boundaries could reflect shared fitting. We therefore repeat the experiment with a confidence field the model supplies itself, with no trained head. We capture the final block's last-token state under a question template, interpolate between matched pairs as before, patch the interpolated state back into the final block, and read the model's own probability of answering true over the two answer tokens. The truth probe is trained as before on a disjoint split of the same states. The endogenous field satisfies the separation premise on a path when it is decisive at both endpoints. Table 3 and Figure 1 report what follows. In Qwen2.5-0.5B the premise holds on 128 of 137 paths, and there the model's own confidence crosses one half within 0.13 of the probe's zero, against null gaps of 0.40 and above. In Pythia-410M the premise fails on every path, since the model answers true regardless of the statement, and the theory predicts nothing. GPT-2 sits between, with 23 of 53 paths decisive and a gap that does not separate from the random-head null. Coupling with the model's own uncertainty appears exactly where the separation premise holds, and only there.

Null baselines. To calibrate the gaps, we replace the trained confidence head with heads trained on permuted labels and with random directions, twenty seeds each (Table 3). The trained-head gaps of 0.25, 0.20, and 0.10 fall below the null medians for all three models, and below the entire null range for Pythia-410M and Qwen2.5-0.5B. For GPT-2 the observed gap lies inside the random-head null range, so the trained-head evidence is weak there on its own.

What this does and does not show. The truth field is linear, so its sign change along a linear path is guaranteed analytically. The empirical content lies elsewhere. The premises hold in real sub-billion models. The two fields are trained on disjoint data with different seeds, so their coupling at the boundary is not an artifact of shared fitting, though both target truth. The endogenous experiment removes that concern wherever the model itself can verbalize truth. The pinned confidence value lands at one half. And the boundary width is a measurable quantity that differentiates models. Code, results, and the figure are included in the supplementary material.

Appendix D. Artifact checklist

The anonymized supplementary zip contains the complete Lean project and the experiment. The Lean part includes the build configuration, the toolchain pin for Lean v4.28.0, the manifest pin for Mathlib v4.28.0, and seventeen source files. To reproduce, run `lake build`. It completes with zero errors and no `sorry`. Axiom sets of the headline theorems are printed by the final-verification file and by the files numbered 13 and 14. The experiment part includes the scripts, the public datasets of Marks and Tegmark (2024), the numerical results, and the figures. The theorems themselves depend on no experiment.

Table 2: Boundary coupling in three sub-1B models. Layer is the selected layer. Probe and Head are held-out accuracies of the truth probe and the confidence head. Verbal is the model's own true or false answer accuracy. Cross is the mean confidence at the probe's zero crossing. Gap is the median distance between the two fields' crossings as a fraction of the path. Width is the relative boundary width.

Model	Layer	Probe	Head	Verbal	Cross	Gap	Width
GPT-2 (124M)	9/12	0.77	0.72	0.48	0.52	0.25	0.33
Pythia-410M	18/24	0.82	0.77	0.50	0.51	0.20	0.36
Qwen2.5-0.5B	12/24	0.96	0.96	0.71	0.49	0.10	0.20

Table 3: The model's own confidence and null calibration. Decisive counts paths where the endogenous field separates the endpoints. Endo is the median crossing gap on decisive paths. Perm and Rand are median null gaps for permuted-label and random-direction heads over twenty seeds. Heads repeats the trained-head gap from Table 2.

Model	Decisive	Endo	Perm	Rand	Heads
GPT-2 (124M)	23/53	0.23	0.48	0.33	0.25
Pythia-410M	0/84	none	0.45	0.45	0.20
Qwen2.5-0.5B	128/137	0.13	0.43	0.40	0.10

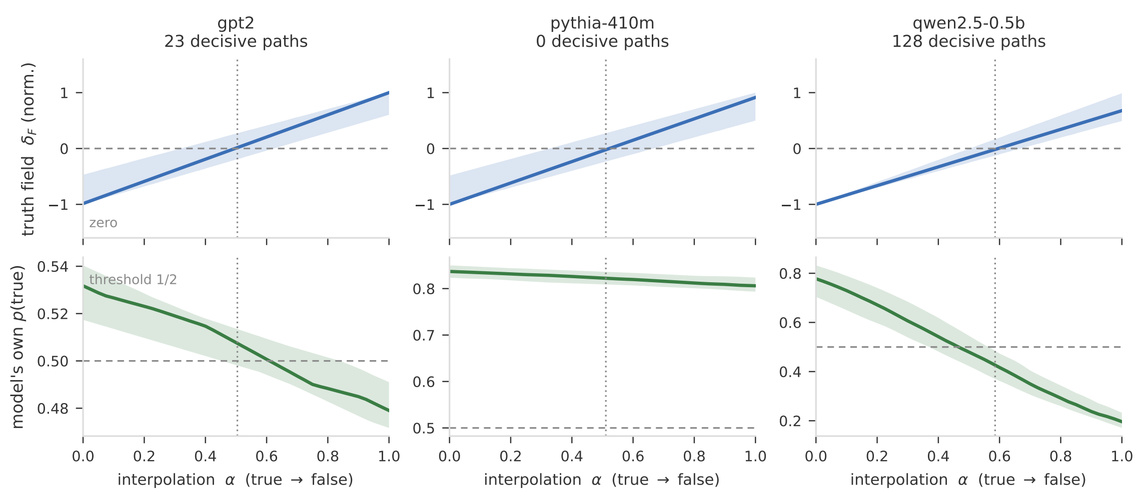

Figure 1: The model's own confidence couples to the probe boundary where the separation premise holds. Rows as in Figure 2, but the bottom row is the model's own probability of answering true, obtained by patching the interpolated final-block state, restricted to decisive paths. Qwen2.5-0.5B sweeps through one half at the probe boundary. Pythia-410M answers true everywhere, the premise fails, and no prediction follows.

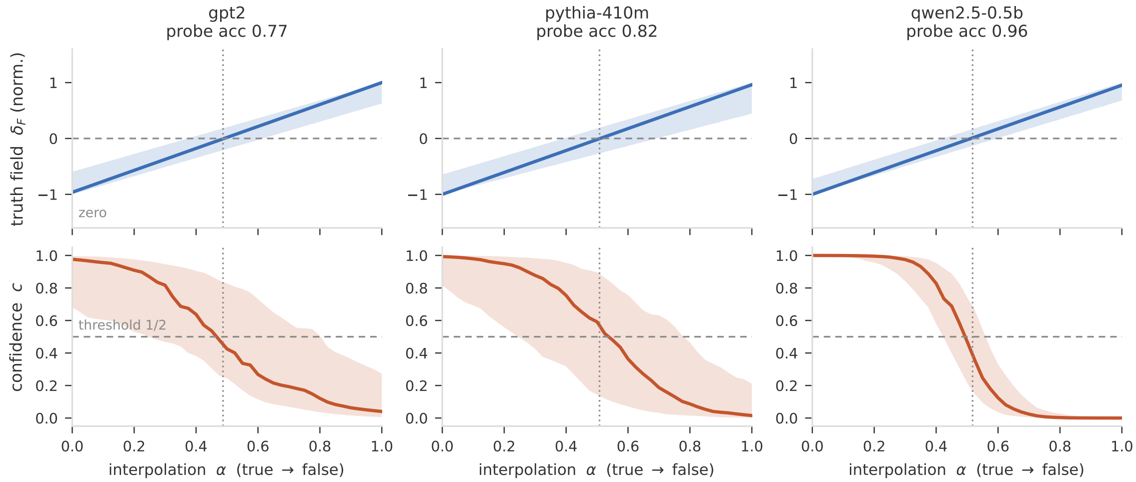

Figure 2: The predicted picture, observed. Top row shows the truth probe field along interpolation paths from a true statement to its negation, median and interquartile band over paths, normalized per path. Bottom row shows the independently trained confidence field on the same paths. The dotted vertical line marks the mean probe zero crossing. Confidence passes through its threshold of one half at the probe boundary in all three models, most sharply in the strongest model.